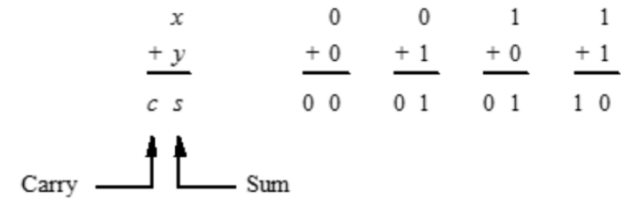
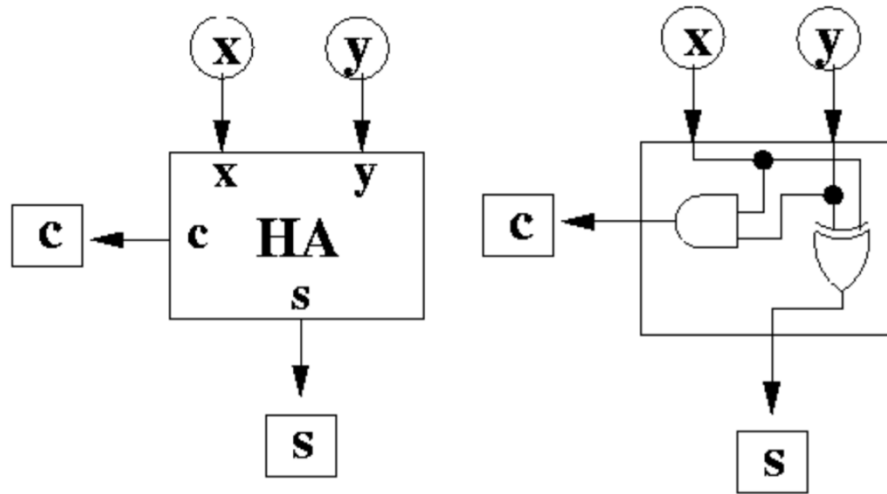


Half Adder

Half Adder

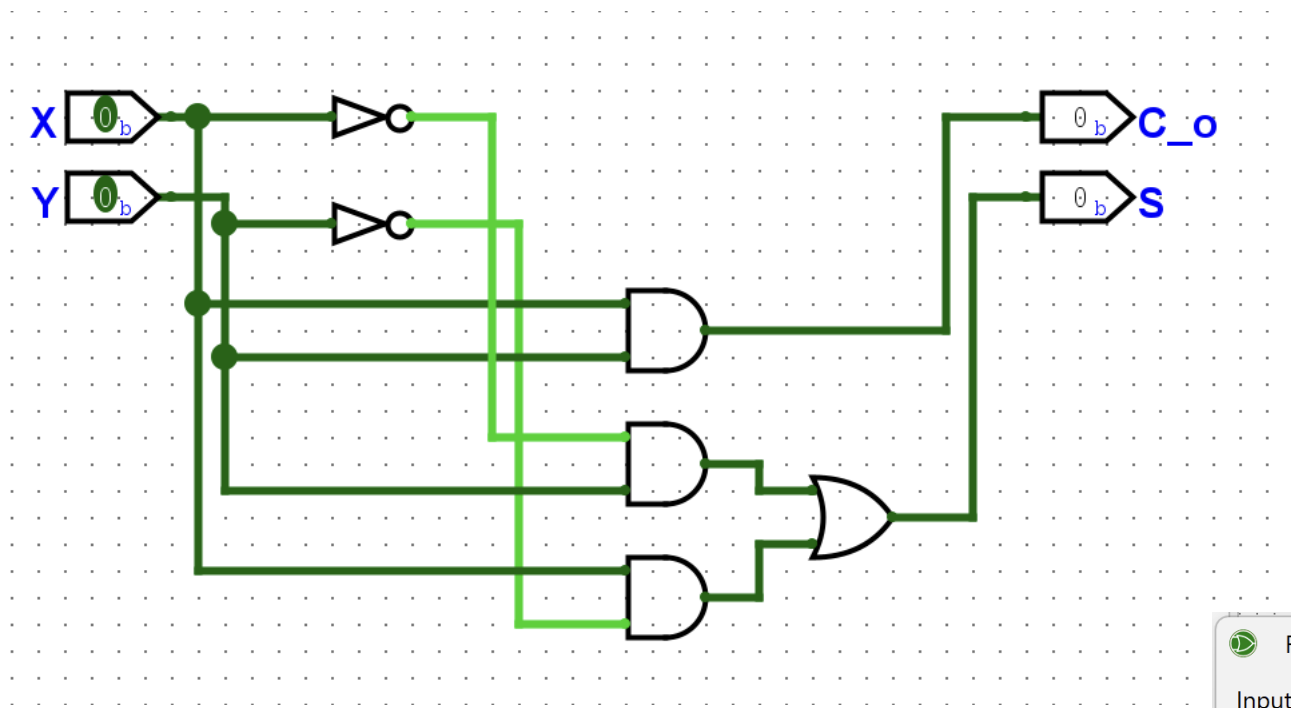
Drawing the circuit:



(a) The four possible cases

x	y	Carry		Sum	
		c		s	
0	0	0		0	
0	1	0		1	
1	0	0		1	
1	1	1		0	

(b) Truth table



File Edit Project Simulate FPGA Window Help Combinational

Inputs & Outputs Table Expression Minimized

- 1 0 Collapse Duplicated Rows Show All Rows

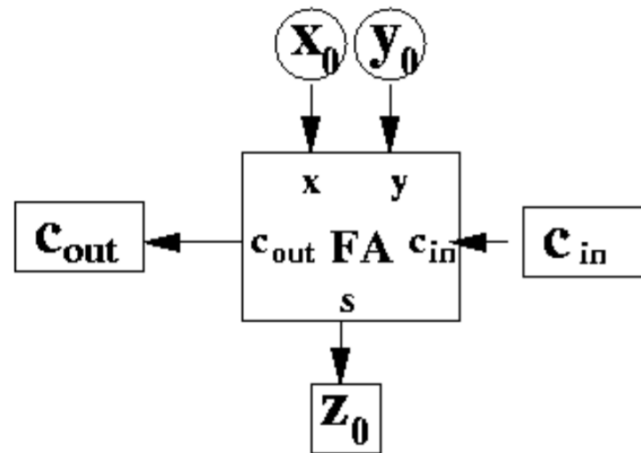
4 of 4 rows shown

X	Y	C_o	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Full Adder

Full Adder

Drawing the circuit:



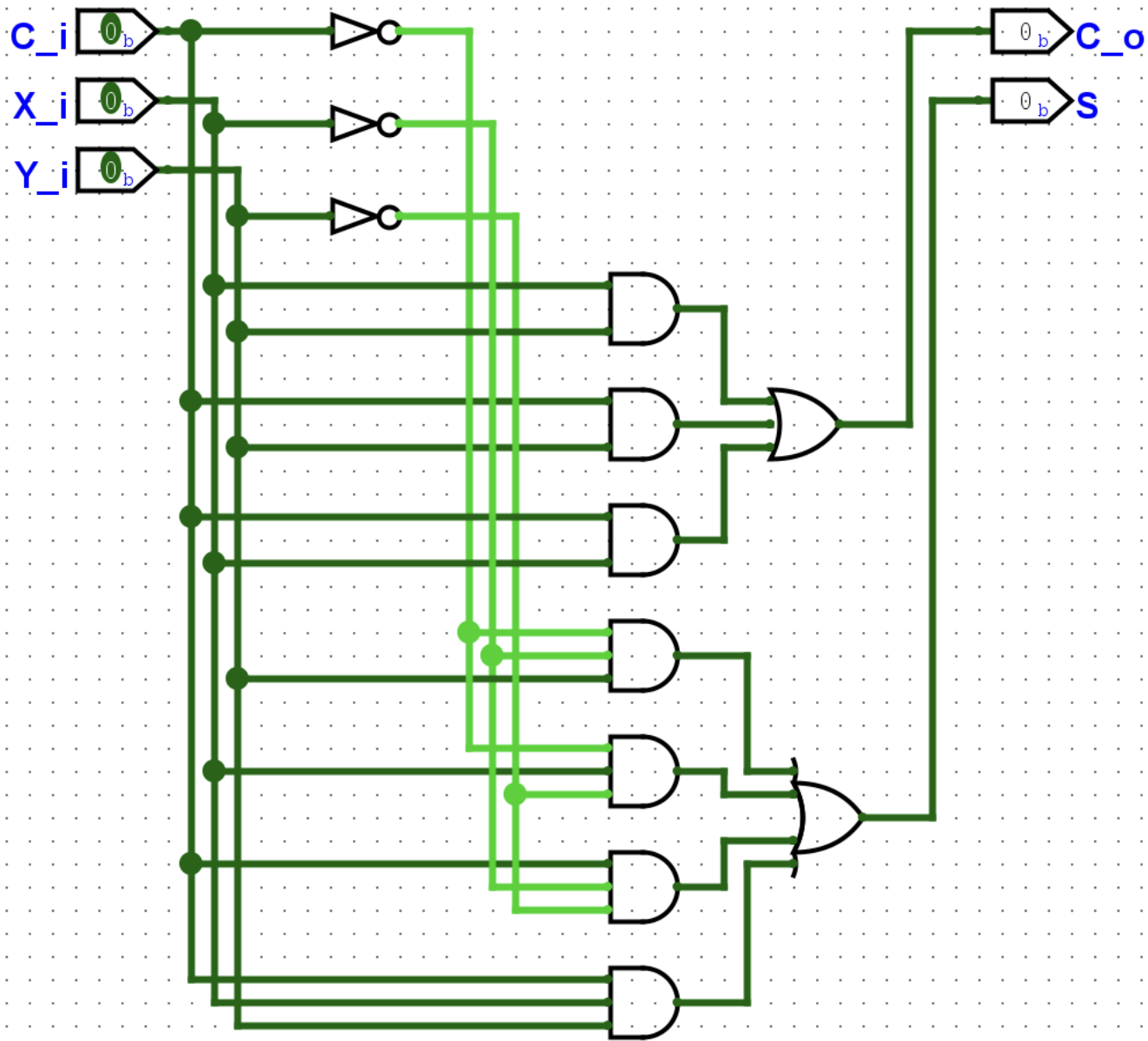
c_i	x_i	y_i	c_{i+1}	s_i
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

(a) Truth table

The diagram shows a hand-drawn addition within a blue rectangular box. It represents the addition of three 1-bit numbers: a carry-in of 1 (in red), and two 1s. The calculation is shown as:
$$\begin{array}{r} 1 \\ + 1 \\ + 1 \\ \hline 11 \end{array}$$

The two 1s being added and the resulting sum '11' are highlighted with a light red background. A blue arrow points from the text 'Carry in from previous column' to the first '1' in the sum.

Carry in
from
previous
column



FPGA Window Help Combinational A

ession Minimized

1 0 Collapse Duplicated Rows Show All Rows | 0

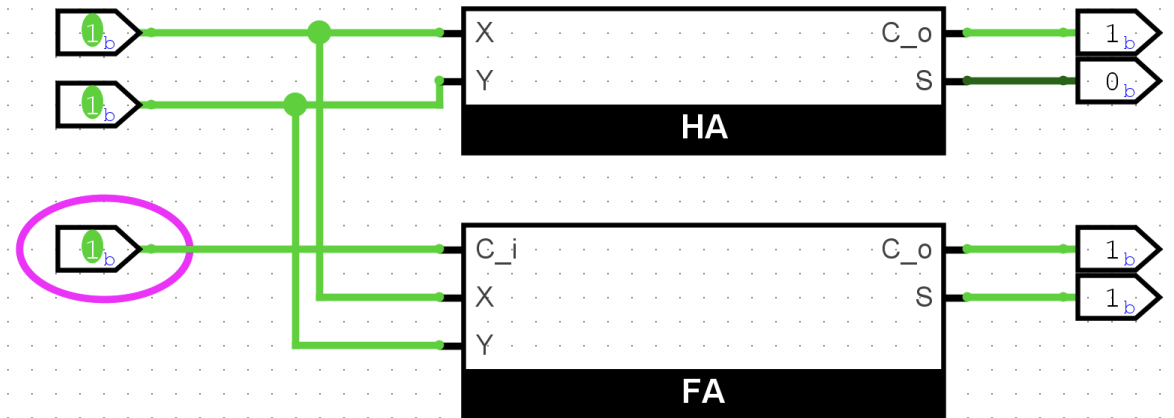
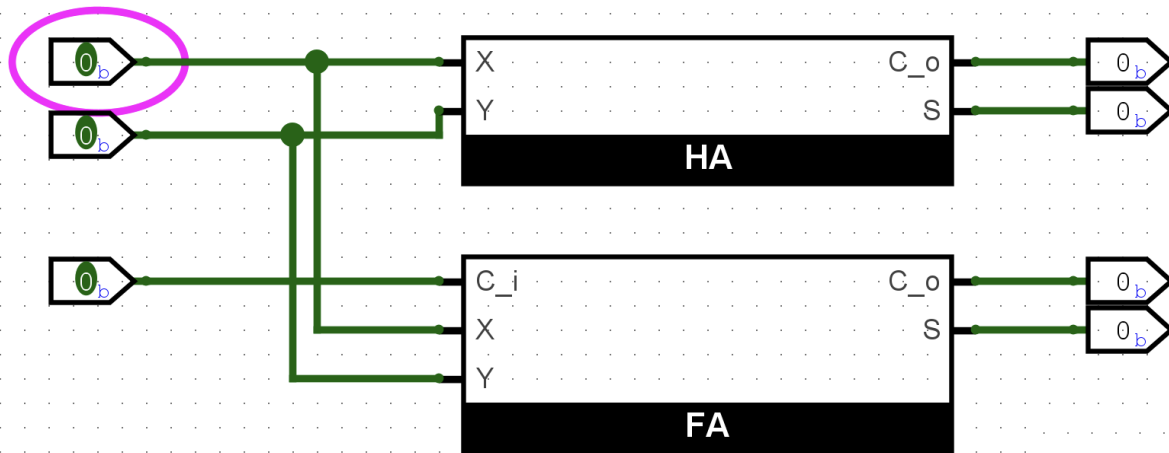
8 of 8 rows shown

C_i	X	Y	C_o	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Tested and Validated

For the HA I am adding $1+1 = 2$

And for the FA I am adding $1 + 1 + 1 = 3$



Ripple Carry

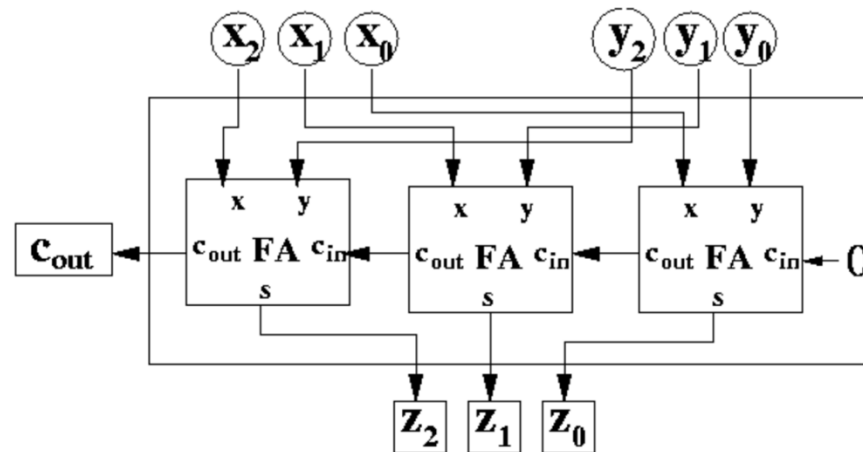
Ripple Carry

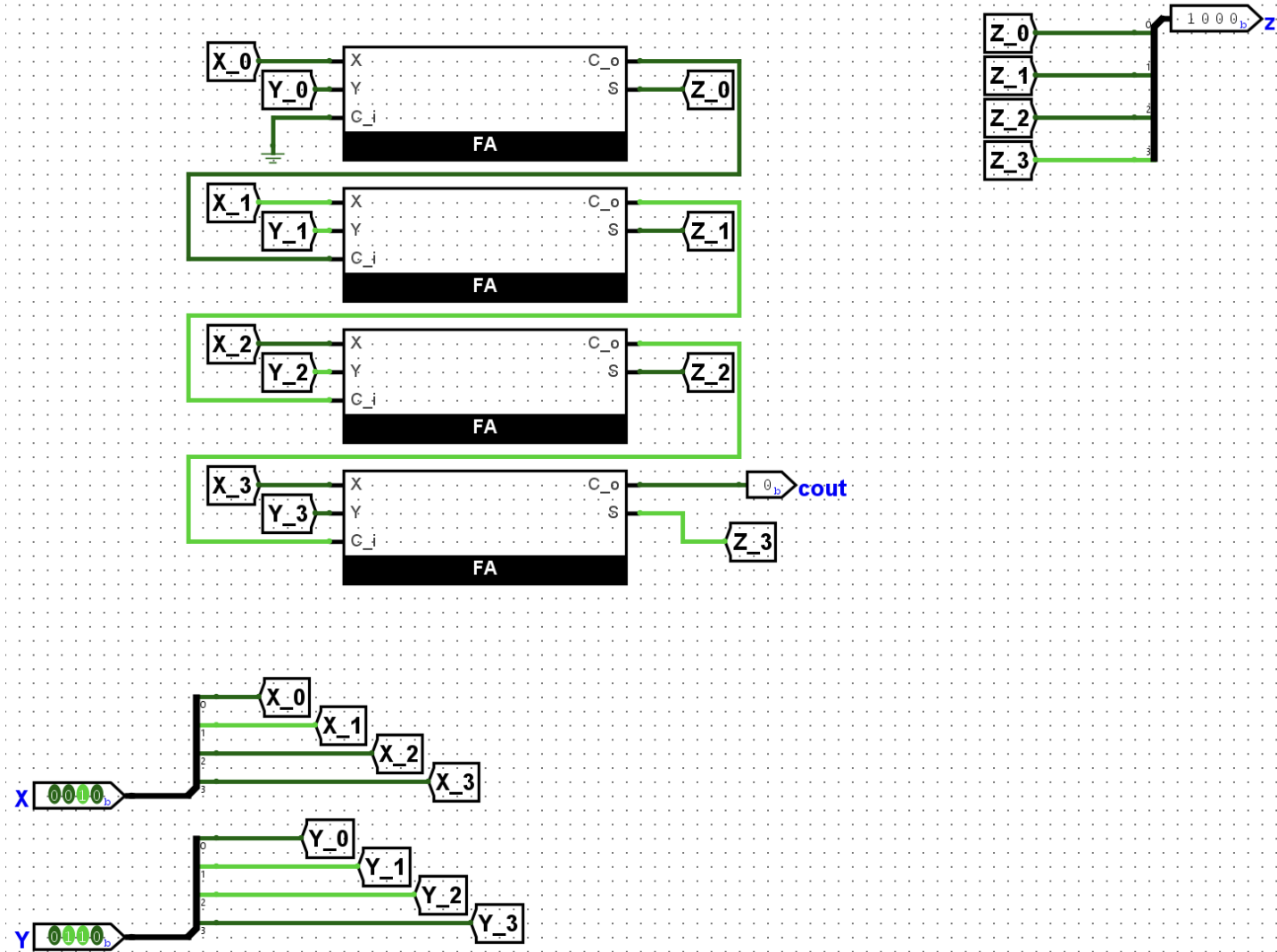
Ripple-Carry Adder

Using only Full Adders:

We don't really need to use the half adder;

We could replace the half adder with a full adder, with a hardwire 0 for the carry in;





Ripple Carry with add and subtract selector

